

Two Structural Obstructions to Galactic Dark Matter in Non-Minimally Coupled Scalar-Tensor Gravity

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Abstract

We prove two structural no-go theorems constraining dark-matter production in scalar-tensor extensions of general relativity with non-minimal coupling $F(\Psi) = \frac{1}{2} + \Lambda_0 \Psi^2$. **Theorem 3 (G_{eff} ceiling):** for any theory with this coupling form, the Solar-System Cassini PPN bound on the Brans–Dicke parameter requires $\Lambda_0 < 1.29 \times 10^{-3} M_{\text{Pl}}^{-2}$, giving a maximum effective gravitational coupling $G_{\text{eff}}/G_N \leq 1 + 2\Lambda_0 \Psi_0^2 \leq 1.018$. Flat galactic rotation curves require $G_{\text{eff}}/G_N \approx 3.1$, which demands $\Lambda_0 \geq 0.154 M_{\text{Pl}}^{-2}$ —incompatible with the Solar-System bound by a factor of ~ 119 . **Theorem 4 (shape inversion):** for any density-dependent coupling $\beta(\rho_b)$ that is monotonically increasing in baryonic density, the predicted gravitational enhancement $E_{\text{model}}(r) = V_{\text{tot}}^2/V_{\text{bary}}^2$ decreases monotonically with galactic radius, while flat rotation curves require an enhancement that increases with radius. The model and requirement are anti-correlated by construction; no parameter choice within the monotone class can produce a positive Pearson coefficient. Numerical evaluation on NGC 3198 gives $r_P = -0.81$ and $\chi^2/\text{dof} = 91.9$. Together the theorems close the two principal modification channels in the quadratic-coupling class: G_{eff} modification and density-dependent fifth-force coupling. We use curvature-memory scalar-tensor gravity (CMSTG) as the worked example and identify precisely to which theories each theorem applies.

Keywords: scalar-tensor gravity, dark matter, modified gravity, post-Newtonian constraints, galactic rotation curves

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1 Introduction

Scalar-tensor extensions of general relativity (GR) provide one of the most systematic frameworks for probing modifications to gravity on astrophysical and cosmological scales [Brans and Dicke \[1961\]](#), [Bergmann \[1968\]](#), [Faraoni \[2004\]](#), [Clifton et al. \[2012\]](#). The canonical non-minimal coupling $F(\Psi)R$, with $F(\Psi) = \frac{1}{2} + \Lambda_0\Psi^2$, is the leading renormalizable departure from GR: the quadratic term is the unique dimension-4 operator available for a real scalar, and it subsumes the Brans–Dicke theory [Brans and Dicke \[1961\]](#), [Damour and Esposito-Farèse \[1992\]](#) as a special case.

Spiral galaxies exhibit flat outer rotation curves that exceed the Newtonian prediction from observed baryonic matter by factors of 2–5 in circular velocity, or 4–25 in gravitational potential [Famaey and McGaugh \[2012\]](#), [McGaugh et al. \[2016\]](#). Within scalar-tensor extensions of GR, two broad mechanism classes address this discrepancy without invoking new matter:

1. *Effective coupling modification* (G_{eff} mechanism): the background scalar field Ψ_0 differs from the field value Ψ_{local} inside a galactic halo, raising the effective Newton constant $G_{\text{eff}} = G_N F(\Psi_0)/F(\Psi_{\text{local}})$.
2. *Density-dependent fifth-force* (chameleon mechanism): the scalar couples to baryonic matter with a coupling $\beta(\rho_b)$ that varies with local density, providing an additional acceleration in the outer disc where baryonic density is low [Khouri and Weltman \[2004a,b\]](#), [Hinterbichler and Khouri \[2010\]](#).

The main results of this paper are:

Theorem 3 (G_{eff} Ceiling). *For any scalar-tensor theory with coupling $F(\Psi) = \frac{1}{2} + \Lambda_0\Psi^2$, the Solar-System Cassini PPN bound requires $\Lambda_0 < 1.29 \times 10^{-3} M_{\text{Pl}}^{-2}$ and caps $G_{\text{eff}}/G_N \leq 1 + 2\Lambda_0\Psi_0^2 \leq 1.018$. The galactic flat-curve requirement $G_{\text{eff}}/G_N \approx 3.1$ demands $\Lambda_0 \geq 0.154 M_{\text{Pl}}^{-2}$. These two constraints have empty intersection; no value of Λ_0 simultaneously satisfies both.*

Theorem 4 (Shape Inversion). *Let $\beta(\rho_b)$ be monotonically increasing in baryonic density ρ_b . Because $\rho_b(r)$ decreases monotonically with galactic radius r outside the disc scale*

length, the fifth-force enhancement $E_{\text{model}}(r) = 1 + 2\beta(\rho_b(r))^2$ is monotonically decreasing with r . Flat rotation curves require $E_{\text{req}}(r) = V_{\text{flat}}^2/V_{\text{bary}}^2(r)$ to be monotonically increasing with r . The Pearson correlation between E_{model} and E_{req} is therefore non-positive for any monotonically increasing $\beta(\rho_b)$ and any galaxy with a falling outer baryonic velocity profile.

We prove both theorems, quantify their scope, and illustrate them with the worked example of curvature-memory scalar-tensor gravity (CMSTG) [Wilson \[2026\]](#), a covariant scalar-tensor theory with action

$$S = \int d^4x \sqrt{-g} \left[\left(\frac{1}{2} + \Lambda_0 \Psi^2 \right) R - \frac{1}{2} (\partial\Psi)^2 - V(\Psi) \right] + S_{\text{SM}}, \quad (1)$$

in units $8\pi G_N = 1$ ($M_{\text{Pl}} = 1$), with canonical locked values $\Lambda_0 = 0.003 M_{\text{Pl}}^{-2}$, $\Psi_0 = 2.62 M_{\text{Pl}}$.

Section 2 proves Theorem 3 and characterises its generality. Section 3 proves Theorem 4 and identifies what it does and does not rule out. Section 4 places the two results in the broader context of dark-matter model-building.

2 Theorem 3: The G_{eff} Ceiling

2.1 Effective gravitational coupling and its maximum

Consider the action (1) with coupling $F(\Psi) = \frac{1}{2} + \Lambda_0 \Psi^2$. Matter is minimally coupled to the Jordan-frame metric $g_{\mu\nu}$, so test masses follow geodesics of $g_{\mu\nu}$. On scales short compared to the inverse scalar mass m_Ψ^{-1} , the effective Newton constant between two test masses depends on the local scalar field value Ψ_{local} :

$$\frac{G_{\text{eff}}}{G_N} = \frac{F(\Psi_0)}{F(\Psi_{\text{local}})} = \frac{\frac{1}{2} + \Lambda_0 \Psi_0^2}{\frac{1}{2} + \Lambda_0 \Psi_{\text{local}}^2}, \quad (2)$$

where Ψ_0 is the cosmological background value. The ratio is maximised in the limit $\Psi_{\text{local}} \rightarrow 0$:

$$\boxed{\frac{G_{\text{eff}}}{G_N} \Big|_{\text{max}} = \frac{\frac{1}{2} + \Lambda_0 \Psi_0^2}{\frac{1}{2}} = 1 + 2\Lambda_0 \Psi_0^2.} \quad (3)$$

This ceiling depends only on Λ_0 and Ψ_0 ; it is independent of the mechanism by which Ψ_{local} is reduced (kernel accumulation, environmental suppression, etc.).

2.2 Brans–Dicke correspondence and the Solar-System bound

The action (1) is equivalent to a Brans–Dicke (BD) theory [Brans and Dicke \[1961\]](#) with $\phi_{\text{BD}} = 2F(\Psi) = 1 + 2\Lambda_0 \Psi^2$. The BD coupling parameter at the cosmological background is [Damour and Esposito-Farèse \[1992\]](#), [Faraoni \[2004\]](#)

$$\omega_{\text{BD}} = \frac{F(\Psi_0)}{(\partial F / \partial \Psi)_{\Psi_0}^2} = \frac{\frac{1}{2} + \Lambda_0 \Psi_0^2}{4\Lambda_0^2 \Psi_0^2}. \quad (4)$$

For a scalar whose Compton wavelength m_Ψ^{-1} exceeds Solar-System scales, the post-Newtonian parameter γ in BD theory is [Will \[2014\]](#)

$$\gamma_{\text{PPN}} = \frac{1 + \omega_{\text{BD}}}{2 + \omega_{\text{BD}}}, \quad |\gamma_{\text{PPN}} - 1| = \frac{1}{2 + \omega_{\text{BD}}}. \quad (5)$$

The Cassini Shapiro-delay measurement [Bertotti et al. \[2003\]](#) gives

$$|\gamma_{\text{PPN}} - 1| < 2.3 \times 10^{-5} \quad \Rightarrow \quad \omega_{\text{BD}} > 43,478. \quad (6)$$

A key prerequisite for mechanism (1) (the G_{eff} modification) to operate at galactic scales ($r \sim 1\text{--}50$ kpc) is that the scalar Compton wavelength must satisfy $m_{\Psi}^{-1} \gtrsim \text{few kpc}$. At these Compton wavelengths, the field is also effectively massless on Solar-System scales ($r_{\text{SS}} \sim 1\text{--}50$ AU $\ll$ kpc), and the PPN formula (5) with constraint (6) applies.

Imposing $\omega_{\text{BD}} > 43,478$ in Eq. (4) and solving for Λ_0 at $\Psi_0 = 2.62 M_{\text{Pl}}$ gives

$$\Lambda_0 < \Lambda_{0,\text{max}} = 1.29 \times 10^{-3} M_{\text{Pl}}^{-2}. \quad (7)$$

The maximum G_{eff} ceiling is then

$$\left. \frac{G_{\text{eff}}}{G_N} \right|_{\text{max}}^{\text{Cassini}} = 1 + 2 \times (1.29 \times 10^{-3}) \times (2.62)^2 \approx 1.018. \quad (8)$$

2.3 Galactic-scale requirement

For a spiral galaxy with flat outer rotation speed V_{flat} and baryonic circular velocity $V_{\text{bary}}(r)$, the G_{eff} -modification hypothesis requires [Famaey and McGaugh \[2012\]](#)

$$\left. \frac{G_{\text{eff}}}{G_N} \right|_{r \gg r_d} = \frac{V_{\text{flat}}^2}{V_{\text{bary}}^2(r)}. \quad (9)$$

For NGC 3198 [Lelli et al. \[2016\]](#) at $r = 30\text{--}44$ kpc: $V_{\text{flat}} \approx 150 \text{ km s}^{-1}$, $V_{\text{bary}} \approx 85 \text{ km s}^{-1}$, giving

$$\left. \frac{G_{\text{eff}}}{G_N} \right|_{\text{required}} = \frac{150^2}{85^2} \approx 3.115. \quad (10)$$

Setting Eq. (3) equal to the galactic requirement yields

$$\Lambda_{0,\text{req}} = \frac{3.115 - 1}{2 \Psi_0^2} = \frac{2.115}{2 \times (2.62)^2} = 0.154 M_{\text{Pl}}^{-2}. \quad (11)$$

2.4 Proof of Theorem 3

Equations (7) and (11) yield:

$$\text{Cassini: } \Lambda_0 < 1.29 \times 10^{-3} M_{\text{Pl}}^{-2}, \quad (12)$$

$$\text{Galactic flat curve: } \Lambda_0 \geq 0.154 M_{\text{Pl}}^{-2}. \quad (13)$$

The incompatibility ratio is $0.154/(1.29 \times 10^{-3}) \approx 119$. The two allowed regions are disjoint; their intersection is empty.

The severity of the violation at $\Lambda_{0,\text{req}}$ confirms this is not a marginal constraint. At $\Lambda_0 = 0.154 M_{\text{Pl}}^{-2}$, the BD parameter (Eq. (4)) drops to $\omega_{\text{BD}} \approx 2.4$, giving $|\gamma_{\text{PPN}} - 1| \approx 0.23$ —exceeding the Cassini limit by a factor of $\sim 10^4$. Five additional precision constraints (CMB temperature spectrum, $f\sigma_8$ at $z < 1$, DESI BAO, UV unitarity, and recombination-era energy budget) fail simultaneously at $\Lambda_{0,\text{req}}$ [Wilson \[2026\]](#).

Theorem 3 is proved. $\square$

2.5 Generality and scope

Higher-order couplings. For $F(\Psi) = \frac{1}{2} + \Lambda_n \Psi^n$ the ceiling becomes $1 + 2\Lambda_n \Psi_0^n$ and the BD parameter scales differently with Λ_n and Ψ_0 , but the same qualitative incompatibility persists for any renormalizable non-minimal coupling: the Cassini bound caps ω_{BD} from below, which caps G_{eff} from above, while the galactic requirement sets a minimum.

Brans–Dicke dual statement. The incompatibility is equivalent to: the Cassini bound requires $\omega_{\text{BD}} > 43,478$, while NGC 3198 requires $\omega_{\text{BD}} \approx 2.4$. No value of ω_{BD} satisfies both.

Screening and massive scalars. A scalar with mass $m_\Psi \gg m_\Psi^{\text{kpc}}$ (Compton wavelength shorter than kpc scales) does not propagate a fifth force at galactic distances and cannot produce the G_{eff} modification of Eq. (2). Such theories evade Theorem 3 automatically because they also cannot explain galactic rotation curves through this mechanism. The theorem applies precisely in the parameter regime where the G_{eff} mechanism is kinematically possible.

What Theorem 3 does not rule out. The theorem is silent on mechanisms that modify the right-hand side of the Einstein equations rather than the left-hand side. Scalar-field dark matter (ultra-light/fuzzy DM, Q-ball condensates, axion halos) contributes through $T_{\mu\nu}$ and is not constrained by the PPN bound on γ . Theories with non-standard conformal factors or disformal couplings require a separate PPN analysis.

CMSTG worked example. At the locked CMSTG values [Wilson \[2026\]](#):

$$\begin{aligned} \omega_{\text{BD}} &= 2,108, & |\gamma_{\text{PPN}} - 1| &= 4.7 \times 10^{-4} & (\text{Cassini: consistent}), \\ G_{\text{eff}}/G_N|_{\text{max}} &= 1.041 & & & (\sim 3\times \text{ short of } 3.115). \end{aligned} \quad (14)$$

A logarithmic sweep of Λ_0 over eight decades (SIM150) numerically confirms the ceiling at every value of Λ_0 consistent with Solar-System and cosmological constraints.

3 Theorem 4: Shape Inversion for Monotone Couplings

3.1 Setup: density-dependent fifth force

A fifth-force modification with density-dependent coupling $\beta(\rho_b)$ to baryonic matter produces an additional acceleration [Khouri and Weltman \[2004a,b\]](#), [Hinterbichler and Khouri \[2010\]](#)

$$a_{5\text{th}}(r) = 2\beta(\rho_b(r))^2 G_N \nabla \Phi_b(r), \quad (15)$$

so that the circular velocity satisfies

$$V_{\text{tot}}^2(r) = V_{\text{bary}}^2(r) [1 + 2\beta(\rho_b(r))^2] \equiv V_{\text{bary}}^2(r) E_{\text{model}}(r). \quad (16)$$

The chameleon and symmetron mechanisms both produce coupling functions $\beta(\rho_b)$ that are monotonically *increasing* with density: large in the dense inner galaxy and in laboratory environments, small in the cosmological void where the scalar is screened.

3.2 Baryon density profile

Outside the stellar disc scale length r_d , the baryonic surface density falls approximately exponentially, $\Sigma_b(r) \propto e^{-r/r_d}$, and the mid-plane volume density $\rho_b(r)$ tracks this fall. Gas discs fall off less steeply, but $\rho_b(r)$ is nonetheless monotonically *decreasing* with r for $r \gtrsim r_d$ in essentially all rotationally supported late-type galaxies.

3.3 The shape-inversion argument

Let $\beta(\rho_b)$ be monotonically increasing. Since $\rho_b(r)$ is monotonically decreasing with r , the composition

$$\beta(r) \equiv \beta(\rho_b(r)) \quad (17)$$

is monotonically *decreasing* with r . Consequently,

$$E_{\text{model}}(r) = 1 + 2\beta(r)^2 \quad \text{is monotonically } \textit{decreasing} \text{ with } r. \quad (18)$$

For a flat rotation curve with $V_{\text{flat}} = \text{const}$ while $V_{\text{bary}}(r) \rightarrow 0$ outside the disc, the required enhancement is

$$E_{\text{req}}(r) = \frac{V_{\text{flat}}^2}{V_{\text{bary}}^2(r)}, \quad (19)$$

which is monotonically *increasing* with r (since $V_{\text{bary}}(r)$ decreases).

A monotone-decreasing function and a monotone-increasing function of the same variable r are anti-correlated. Their Pearson correlation coefficient satisfies $r_P \leq 0$, with equality only if one is constant. No choice of $(\beta_0, \rho_{\text{sc}})$ within the monotonically increasing class can produce $r_P > 0$.

Theorem 4 is proved. $\square$

3.4 Numerical confirmation on NGC 3198

Table 1 shows E_{req} and E_{model} at four reference radii for NGC 3198 ($V_{\text{flat}} = 150 \text{ km s}^{-1}$, $r_d = 2.7 \text{ kpc}$) Lelli et al. [2016], using the tanh profile $\beta(\rho_b) = \beta_0 \tanh^2(\rho_b/\rho_{\text{sc}})$ with self-consistent coupling $\beta_0 = 1.018$ Wilson [2026].

r [kpc]	2	10	20	40
E_{req}	1.80	2.60	3.80	5.20
E_{model}	3.10	1.07	1.04	1.04

Table 1: Required and predicted enhancement factors for NGC 3198. E_{model} peaks in the inner disc (high density, large β) and saturates near unity in the outer disc; E_{req} rises monotonically outward. The opposite trend in each quantity is the structural anti-correlation of Theorem 4.

The Pearson correlation between E_{req} and E_{model} across all 43 SPARC data points is $r_P = -0.81$, and the best-fit $\chi^2/\text{dof} = 91.9$ against an acceptance threshold of ≤ 2.0 . The optimal transition density ρ_{sc} is pinned at its lower boundary, effectively the uniform limit in which chameleon variation is minimised—the mechanism is most effective when it varies least, which is a restatement of the anti-correlation. These results are reproduced analytically from the monotonicity argument above.

3.5 Generality and scope

Galaxy and profile independence. The proof requires only that $\rho_b(r)$ is monotonically decreasing and $\beta(\rho_b)$ is monotonically increasing, both of which hold universally outside the disc scale length. The anti-correlation and negative Pearson coefficient are therefore properties of the entire class of monotonically increasing $\beta(\rho_b)$, not of any specific galaxy or profile parameter.

What Theorem 4 does not rule out.

1. *Monotonically decreasing $\beta(\rho_b)$* (“anti-chameleon”): a coupling that decreases with density gives $\beta(r)$ increasing with r , producing $E_{\text{model}}(r)$ increasing with r . Theorem 4 is evaded. However, anti-chameleon profiles face an independent obstruction: at cosmological densities $\rho_b \sim \rho_{\text{crit}}$, a decreasing-with-density coupling is large (unscreened), generically violating CMB and BAO constraints at order unity. A self-consistent CMSTG analysis constrains the cosmological coupling to $\beta_\infty \lesssim 2.4 \times 10^{-10}$ Wilson [2026], which rules out anti-chameleon solutions for that theory.
2. *Non-monotone $\beta(\rho_b)$* : a coupling that peaks at intermediate galactic densities and is small at both stellar-core and cosmological densities can, in principle, produce E_{model} peaking in the outer disc. Theorem 4 does not exclude this. The standard thin-shell chameleon always produces monotone $\beta(\rho_b)$ in its usual operating regime; non-monotone profiles require a qualitatively different mechanism architecture.
3. Mechanisms operating through $T_{\mu\nu}$ sourcing (scalar dark matter, condensates, ultra-light fields): these modify the right-hand side of the Einstein equations and are not constrained by the fifth-force formula (15).
4. Coupling functions depending on variables other than ρ_b —for example, the tidal field, temperature, or ionization fraction. For these, the argument that $\beta(r)$ is decreasing requires additional galaxy-specific input.

CMSTG worked example. In CMSTG, the coupled chameleon coupling is $\beta(\rho_b) = \beta_0 \tanh^2(\rho_b/\rho_{\text{sc}})$ with $\beta_0 = 1.018$ set by cosmological screening Wilson [2026]. The full SPARC rotation-curve test (SIM152) evaluates this against 43 NGC 3198 data points, confirming $\chi^2/\text{dof} = 91.9$ and $r_P = -0.81$.

4 Discussion and Conclusions

We have proved two structural no-go theorems for dark-matter production in scalar-tensor theories with non-minimal coupling $F(\Psi) = \frac{1}{2} + \Lambda_0 \Psi^2$.

Theorem 3 is a *numerical* incompatibility between two specific observational requirements. Its conclusion (Λ_0 incompatible by factor 119) is specific to the combination $|\gamma_{\text{PPN}} - 1|_{\text{Cassini}}$, V_{flat} and V_{bary} for NGC 3198, and Ψ_0 . A researcher applying Theorem 3 to a different scalar-tensor theory must re-derive the analog of Eqs. (12)–(13) for their own field value Ψ_0 and coupling Λ_0 . The qualitative incompatibility is expected to persist for any quadratic non-minimal coupling.

Theorem 4 is a *geometric* (anti-monotone) incompatibility. It requires no numerical inputs beyond the fact that $\rho_b(r)$ decreases and V_{flat} is roughly constant in the outer disc.

The negative Pearson coefficient is guaranteed for any monotonically increasing $\beta(\rho_b)$ and any galaxy with a falling baryonic velocity profile. It generalizes to other theoretical frameworks without re-derivation.

The contrast in character is relevant for future researchers choosing between the two theorems. Theorem 3 is the sharper tool when comparing the G_{eff} mechanism against the specific observational inputs needed for the target theory; Theorem 4 is the sharper tool for ruling out entire classes of monotone couplings without galaxy-specific computation.

Taken together, the theorems establish that any viable dark-matter mechanism within the $F(\Psi) = \frac{1}{2} + \Lambda_0 \Psi^2$ coupling class must either:

- (a) Operate through the energy-momentum $T_{\mu\nu}$ channel rather than through a modification of G_N or a density-dependent fifth force, or
- (b) Use a non-monotone coupling $\beta(\rho_b)$ with independent theoretical motivation beyond the standard chameleon construction.

The theorems do not rule out scalar-tensor theories as hosts for dark matter; they sharpen which *mechanism* can accomplish this.

For CMSTG specifically, the two theorems (together with the mass-gap obstruction for the sextic-condensate channel documented in Wilson [2026]) exhaust the three available extension channels for dark-matter production within the locked Phase 1 action. The implication is that if CMSTG correctly describes the large-scale universe, dark matter in the observed universe must come from a sector structurally distinct from the scalar-tensor framework—standard candidates (cold dark matter, axions, sterile neutrinos) remain fully compatible with CMSTG’s cosmological successes.

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